

LoRA + DoRA as Extension: Weight-Decomposed Low-Rank Adaptation on GPT-2

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Papers:
Hu et al. LoRA, ICLR 2022
Liu et al. DoRA, ICML 2024

Primary: LoRA | Extension: DoRA

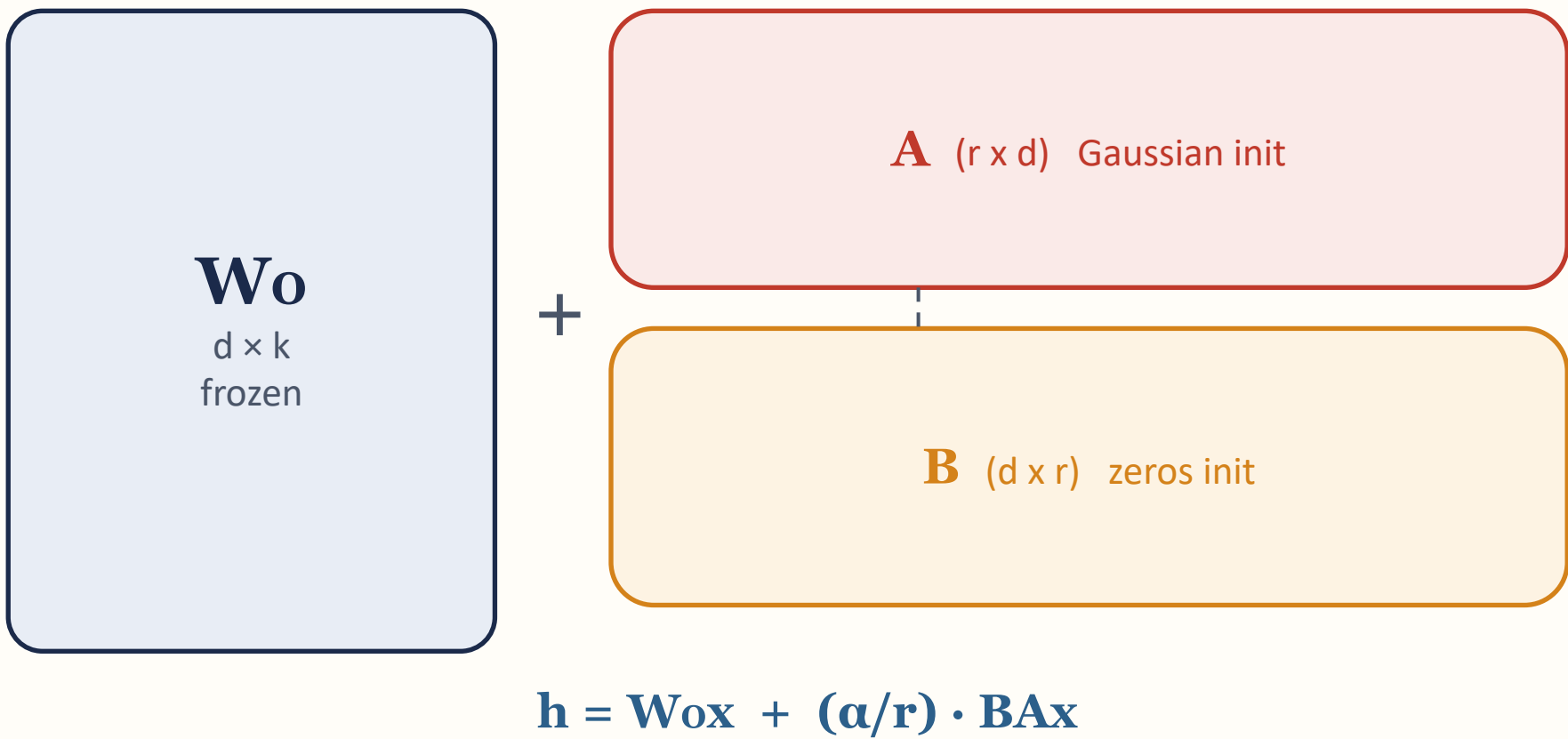
The Problem

- Full fine-tuning updates all 124M params

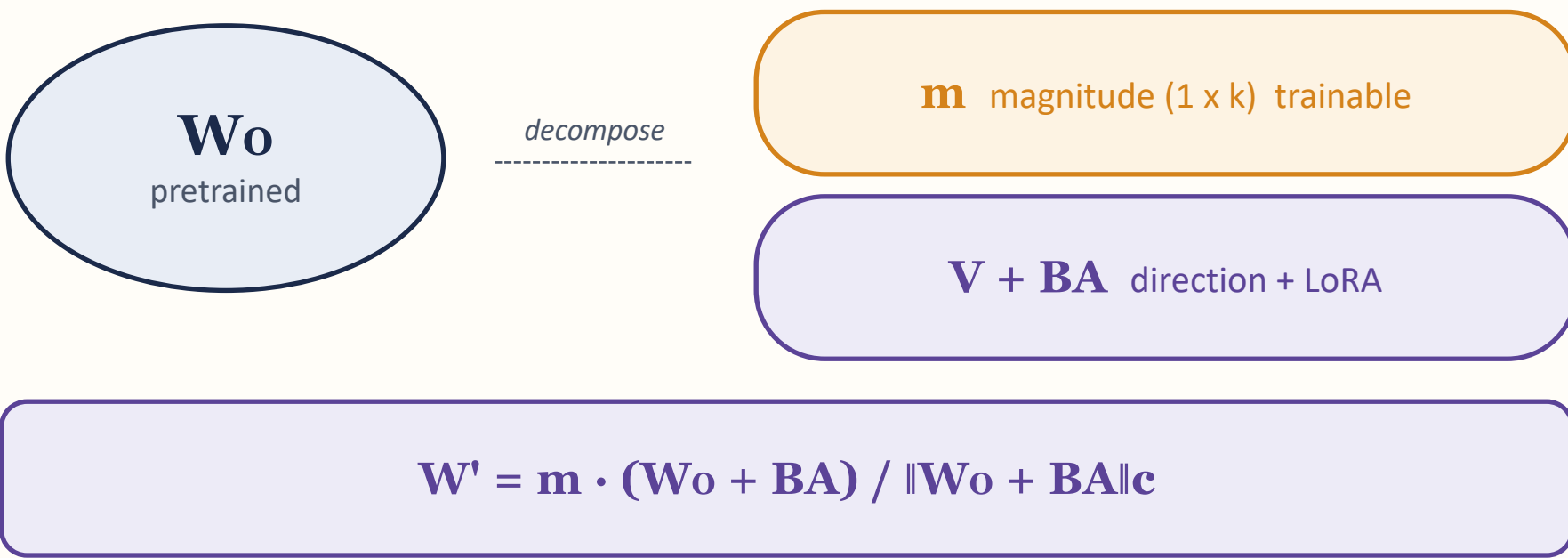
- Adam optimizer state alone: ~936 MB

Hypothesis: LoRA matches near-full-FT perplexity with far fewer params. DoRA further improves over LoRA at the same rank.

How LoRA Works



DoRA's Key Insight



WikiText-2 — sample from training data

Eiffel Tower

The tower is 324 metres (1,063 ft) tall, an aerial is situated on the top of the tower and its base is in the field ; its total height is 330 metres (1,083 ft). As tall as an 81-storey building, the Eiffel Tower is the tallest structure in Paris and was the tallest man-made structure in the world from its completion in 1889 until the Chrysler Building in New York City was built in 1930 ...

Model sees tokenized input:

The [tower] is [324] metres [tall] [an] aerial ...

Each token is predicted from context — model learns language structure

Correlation Analysis

Full FT	LoRA	DoRA
-0.62	+0.83	-0.31
Decoupled (as expected)	Coupled (limited)	Closer to full FT

Experimental Setup

Model: GPT-2 small (117M) **Data:** WikiText-2 causal LM **Metric:** Validation perplexity **No PEFT library**

Validation Perplexity (lower = better)

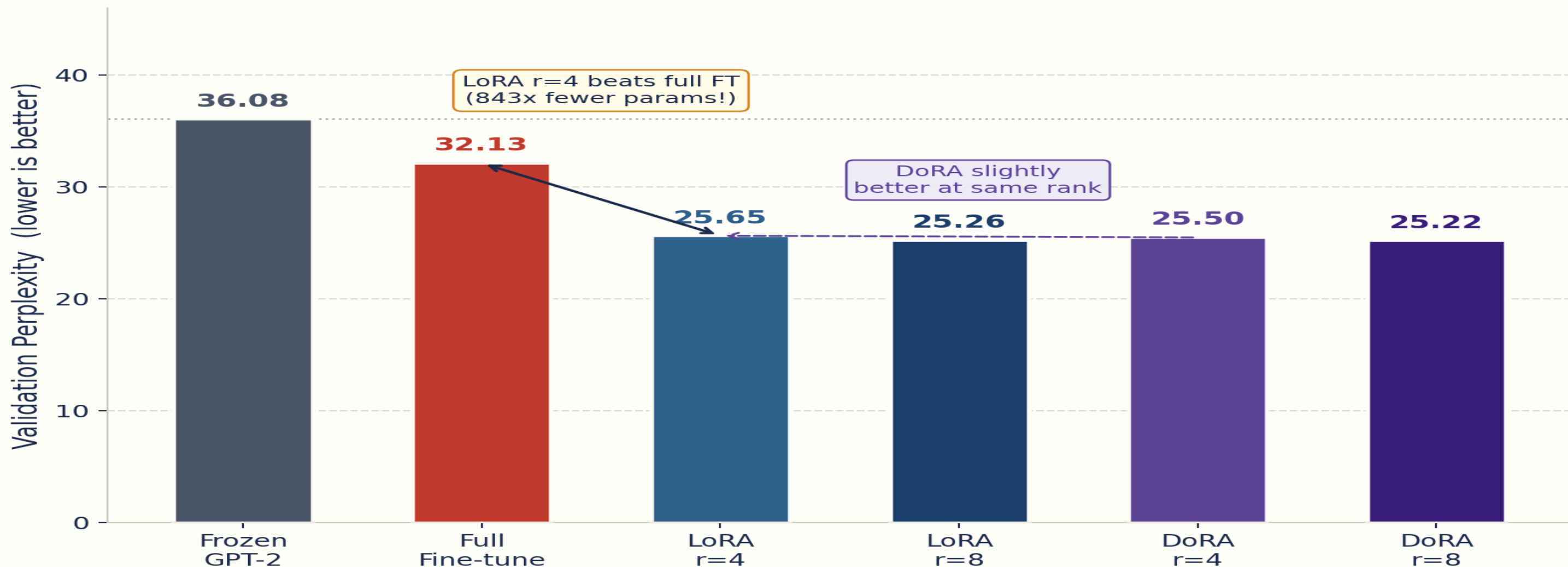


Fig 1. All adapters beat full fine-tuning. LoRA's low-rank constraint prevents overfitting. Full FT overfits after epoch 1. DoRA consistently outperforms LoRA at every rank.

Parameter Efficiency + Memory

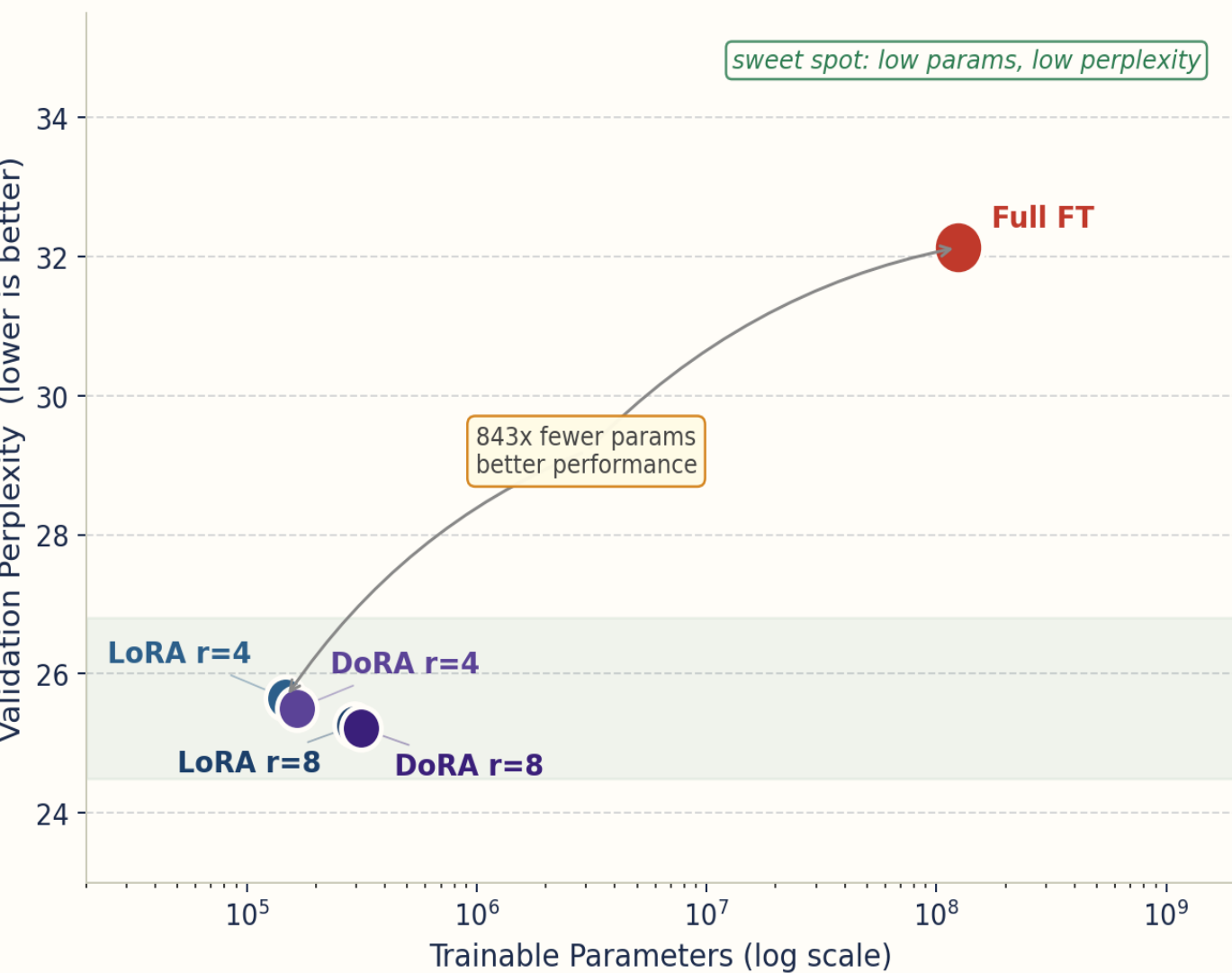


Fig 2. Low params, low perplexity -> PEFT hit sweet spot on WikiText-2.

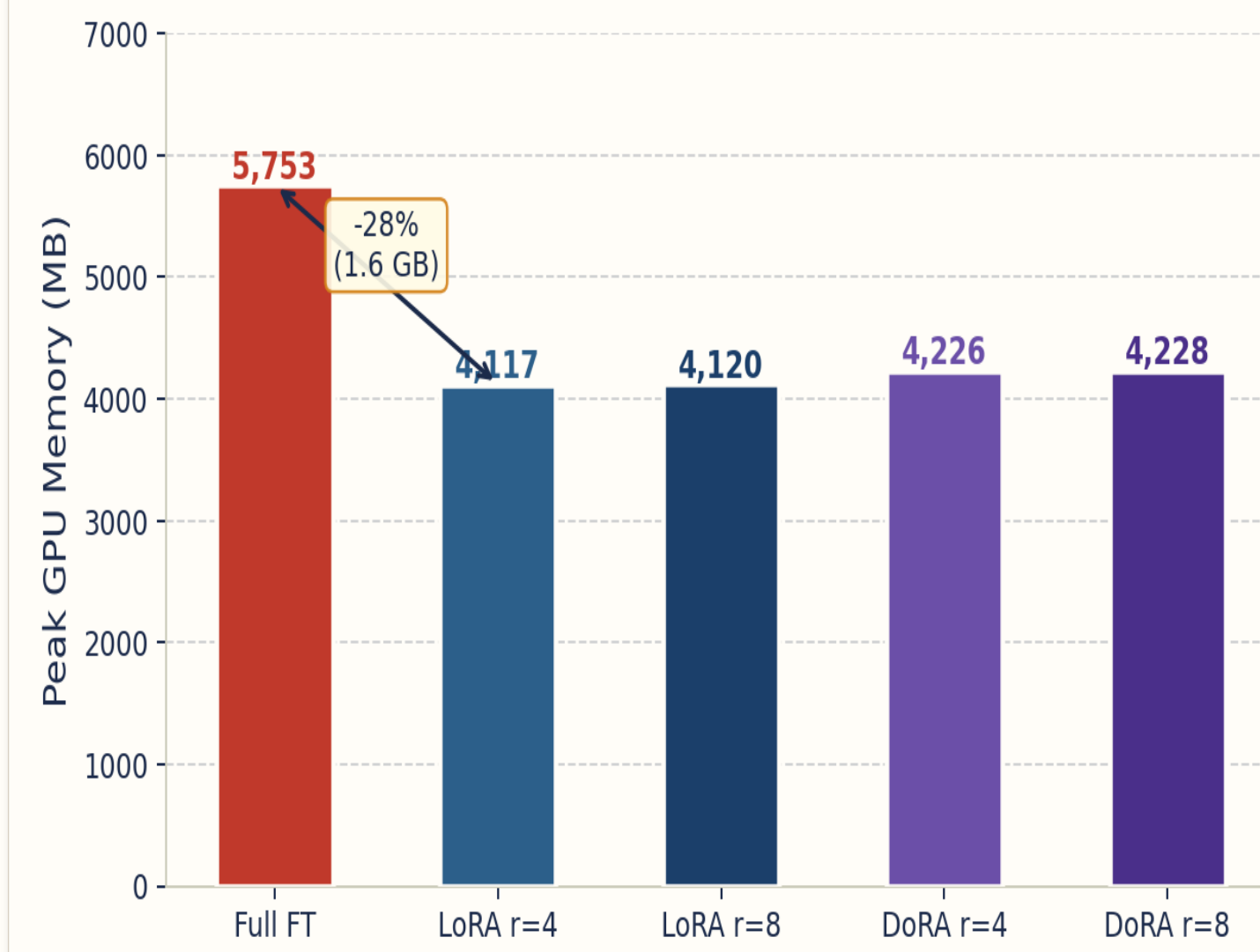


Fig 3. LoRA saves 1.6 GB vs full FT. DoRA adds only ~110 MB.

Training Curves

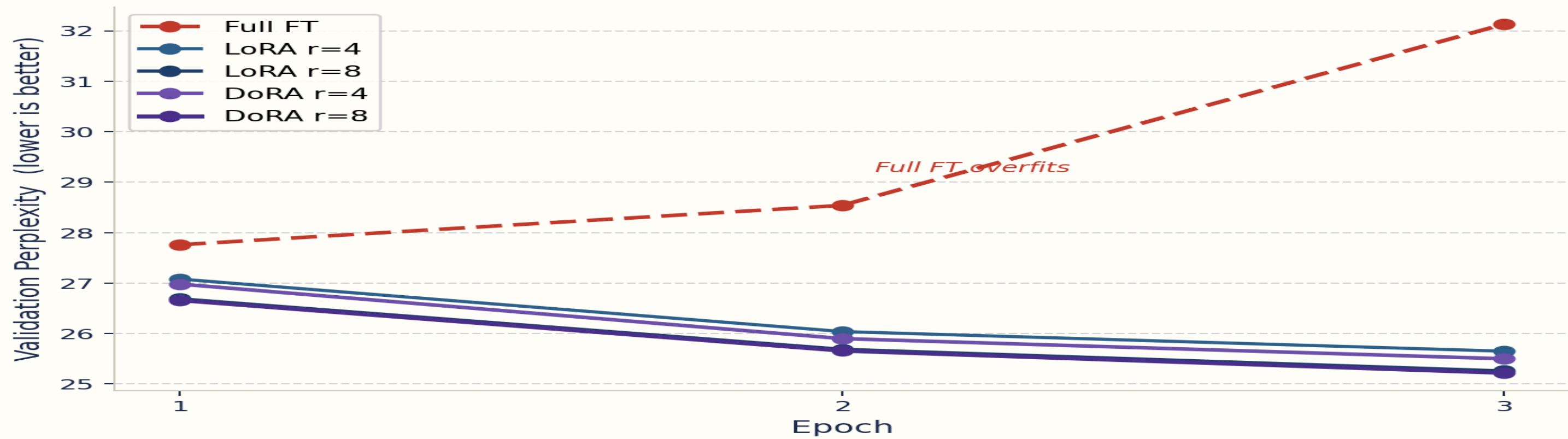


Fig 4. Full FT perplexity rises after epoch 1 (overfitting). LoRA and DoRA converge cleanly. DoRA r=8 achieves best result: 25.22.

Results Summary

Configuration	Params	Val PPL	GPU Mem
Frozen GPT-2	0	36.08	--
Full Fine-tune	124M	32.13	5,753 MB
LoRA r=4	147K	25.65	4,118 MB
LoRA r=8	295K	25.26	4,120 MB
DoRA r=4	166K	25.50	4,227 MB
DoRA r=8 *	313K	25.22	4,228 MB

DoRA rows = my extension

Key Findings

- Unexpected:** LoRA beats full FT (25.65 vs 32.13). Low-rank acts as implicit regularization.
- DoRA wins:** Better perplexity across all tested ranks for only ~12% more trainable params.
- Memory:** LoRA saves 1.6 GB vs full FT. DoRA adds only ~110 MB over LoRA.
- Scale matters:** DoRA gap over LoRA likely grows with larger models and harder tasks.

Takeaways

PEFT works: 843x fewer params, better results

DoRA = free upgrade: similar params, consistently better

Future Work

Apply DoRA to FFN layers (not just Q, V)

Test DoRA+ (half-rank): same params as LoRA r=4, better accuracy?

Scale to GPT-2 medium/large to test if DoRA gap widens